

Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL LEAD-FREE

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BUYER.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE

BOARD THICKNESS

TRACE WIDTH

TRACE TO TRACE

MIN. HOLE (PTH)

MIN. HOLE (NPTH)

ANNULAR RING

COPPER TO HOLE

COPPER TO EDGE

HOLE TO HOLE

85,000 × 100,000 mm

1,600 mm

0,300 mm

0,020 mm

0,250 mm

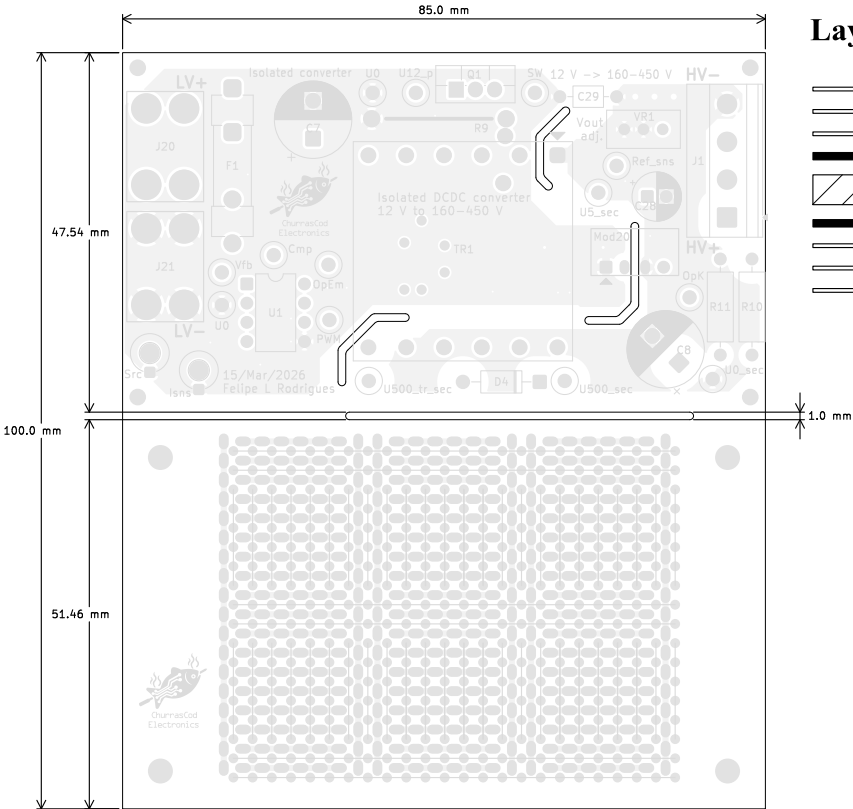
2,100 mm

0,150 mm

0,200 mm

0,250 mm

0,254 mm



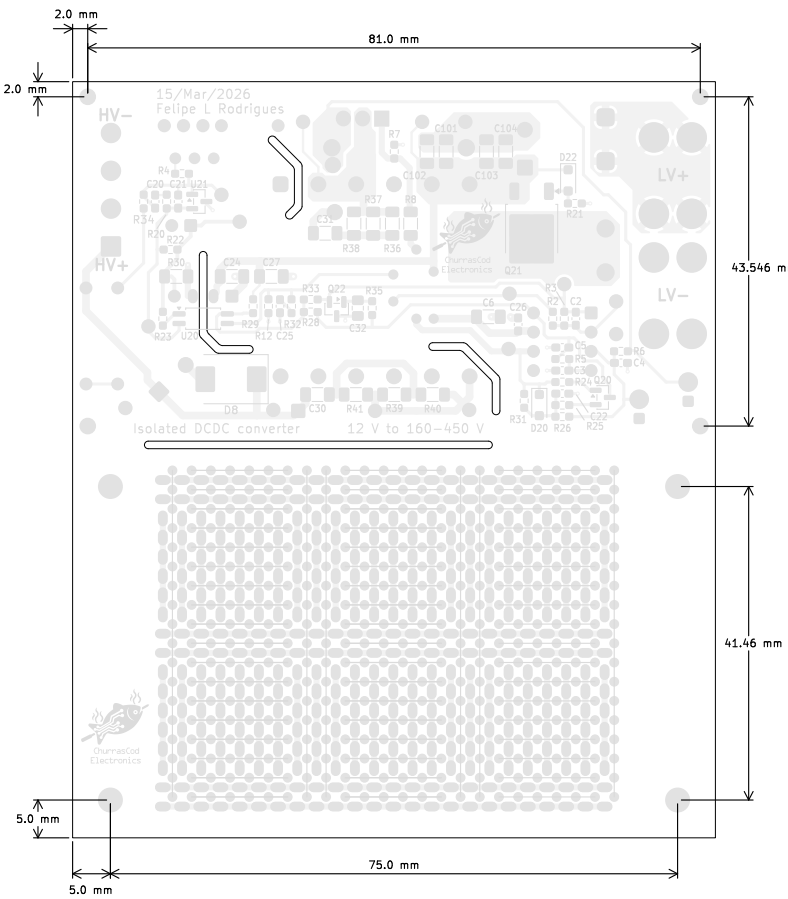
Layer Stack Legend

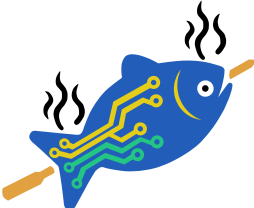
Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen			Direct Printing	Legend	GBR
F.Mask		0,02 mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
Core		1,49 mm	FR4	Dielectric	
Copper	L2 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
B.Mask		0,02 mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,6 mm
Note: external layer thicknesses are specified after plating



Comments:	Company: NA		Variant: CHECKED		Git Hash: 85c4a65	
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V			
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb		Designer: FR		Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:			Reviewer: FR		Size: A4	Sheet: 1 of 10



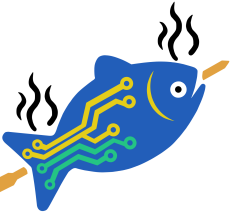
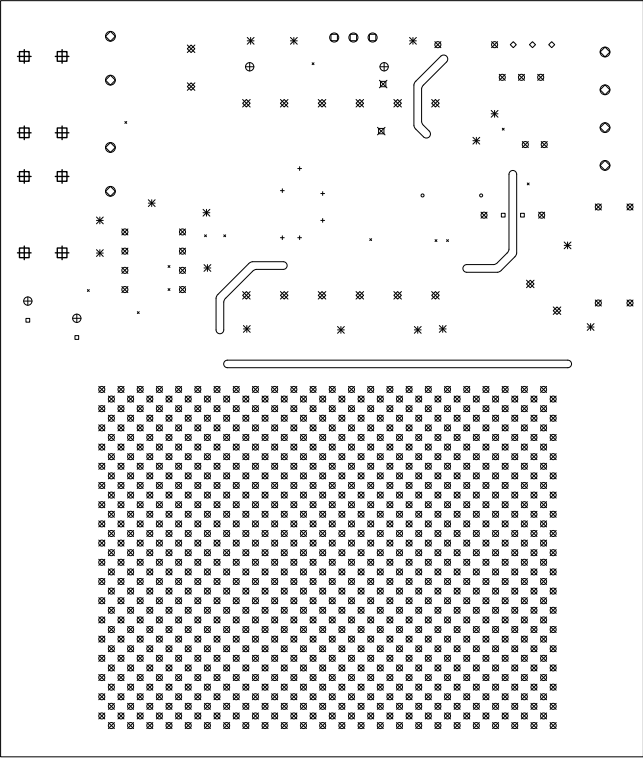
 ChurrasCod Electronics	Comments:		Company: NA		Variant: CHECKED	Git Hash: 85c4a65
	Sheet Title: Bottom Fabrication (Scale 1:1)		Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
	Sheet Path:		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
			Reviewer: FR	Size: A4	Sheet: 2 of 10	

Step-Up Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0.25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
○	2	0.40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
+	6	0.50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
□	4	0.70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
◇	3	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
⊗	885	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
✱	16	0.90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊗	2	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
⊗	16	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	4	1.10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	3	1.20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
○	8	1.30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	8	1.80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
Total 970						



ChurrasCod
Electronics

Comments:

Company:

NA

Variant:

CHECKED

Git Hash:

85c4a65

Board Name:

Step-Up

Project Name:

Isolated step-up 12:450 V

Sheet Title:

Drill Drawing (L1 - L4)

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Date:

2026-02-07

Revision:

2.0.0+ (Unreleased)

Sheet Path:

Reviewer:

FR

Size:

A4

Sheet:

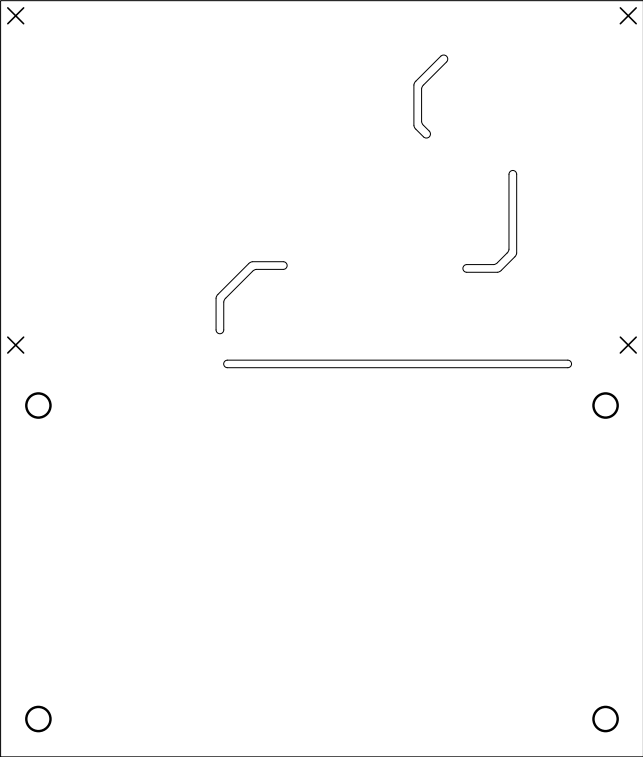
3 of 10

Step-Up Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

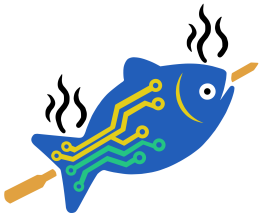
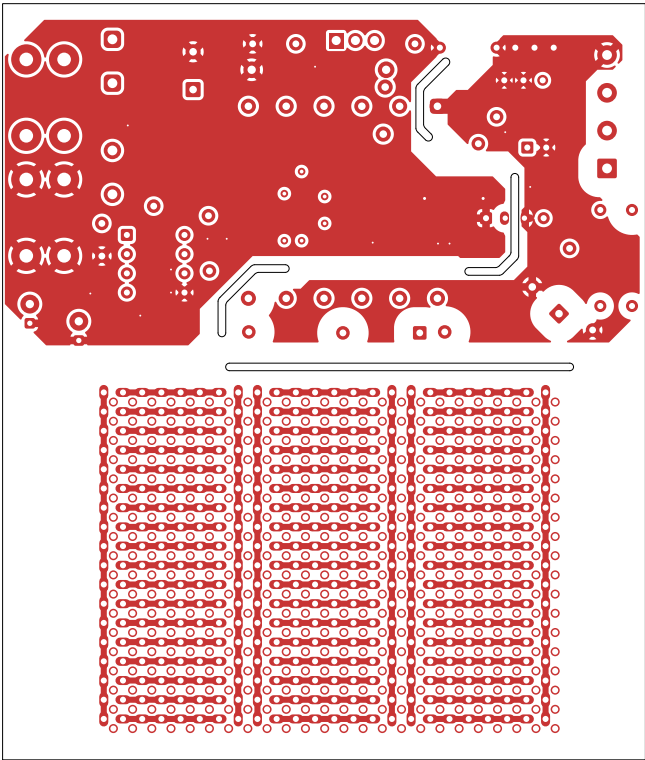
Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
○	4	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
Total 8						



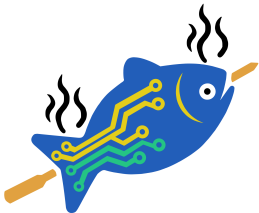
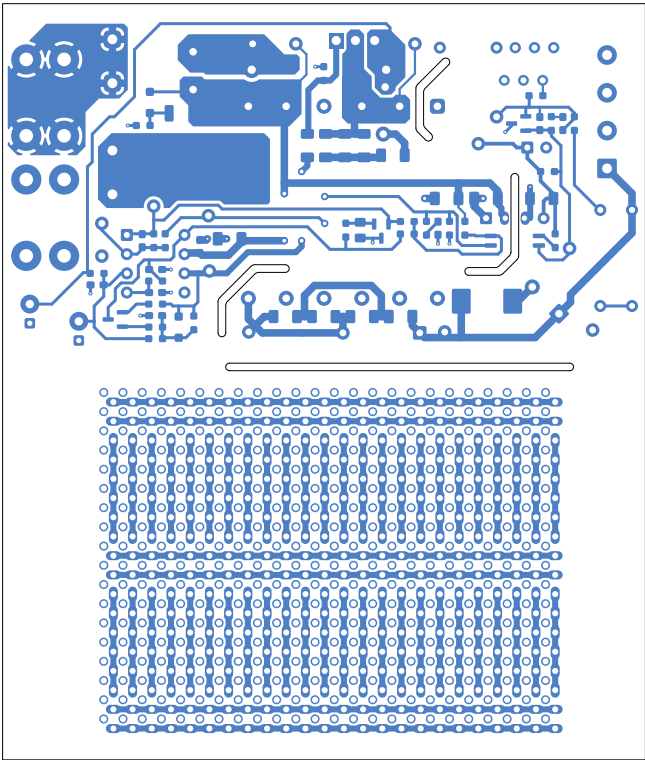
ChurrasCod
Electronics

Comments:	Company: NA		Variant: CHECKED	Git Hash: 85c4a65	
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V		
Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)	
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 4 of 10	



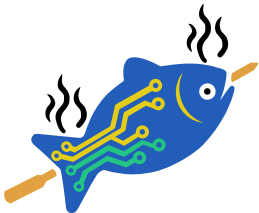
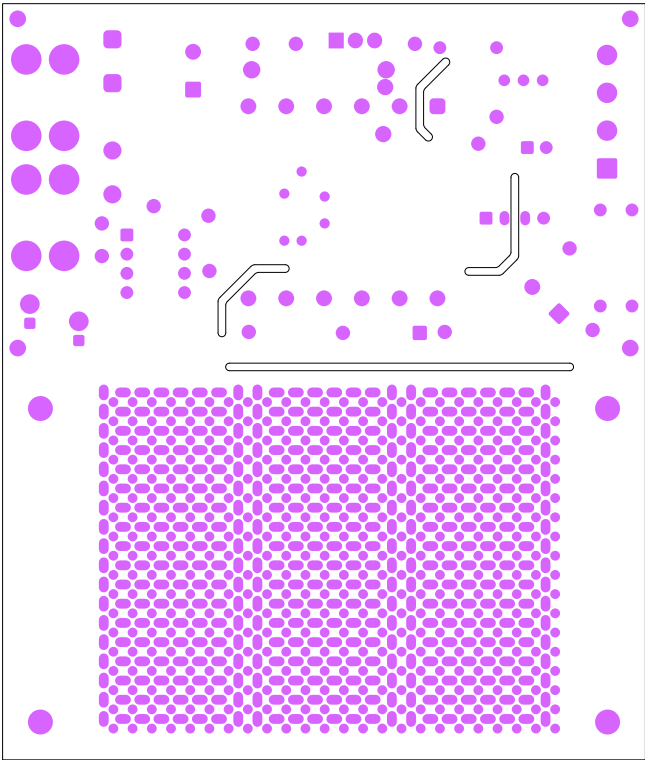
ChurrasCod
Electronics

Comments:	Company:		Variant:	Git Hash:
	NA		CHECKED	85c4a65
	Board Name:		Project Name:	
Sheet Title:	File Name:	Designer:	Date:	Revision:
F.Cu (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-07	2.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size:	Sheet:
		FR	A4	5 of 10

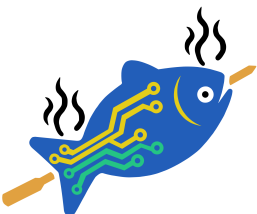


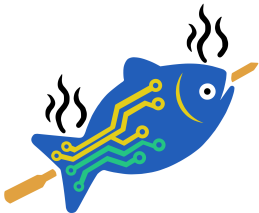
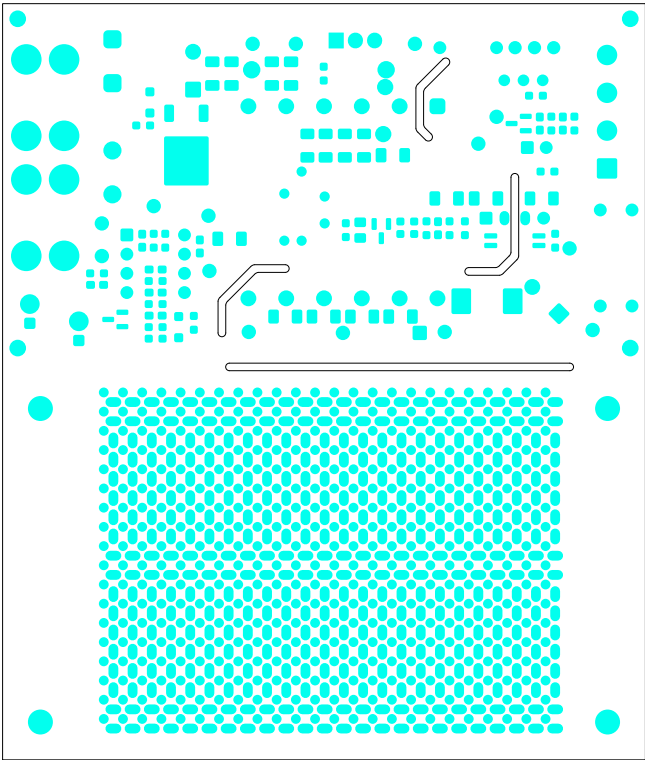
ChurrasCod
Electronics

Comments:	Company: NA		Variant: CHECKED	Git Hash: 85c4a65
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: B.Cu (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 6 of 10

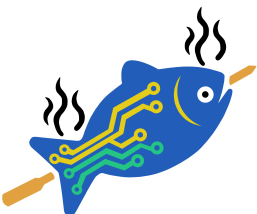


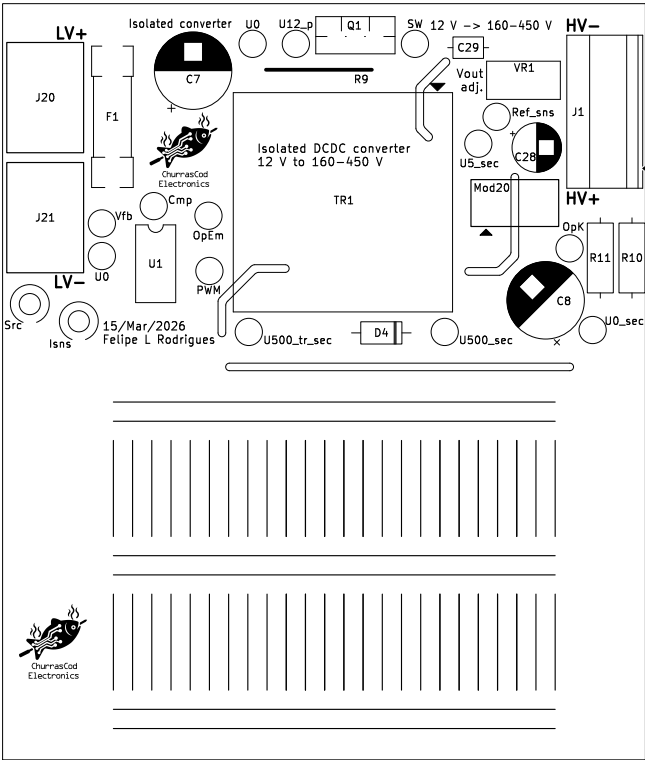
ChurrasCod
Electronics

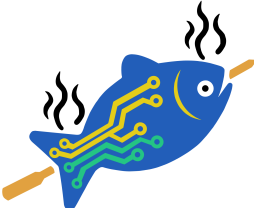
 ChurrasCod Electronics	Comments:		Company:		Variant:	Git Hash:
			NA		CHECKED	85c4a65
	Board Name:				Project Name:	
	Step-Up				Isolated step-up 12:450 V	
Sheet Title:	File Name:	Designer:	Date:	Revision:		
F.Mask (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-07	2.0.0+ (Unreleased)		
Sheet Path:		Reviewer:	Size:	Sheet:		
		FR	A4	7 of 10		

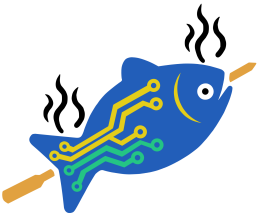
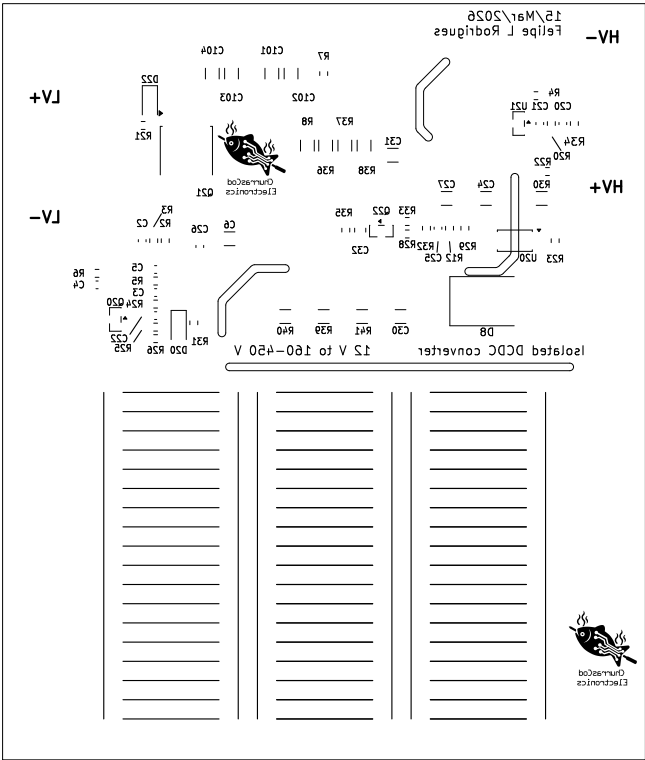


ChurrasCod
Electronics

 ChurrasCod Electronics	Comments:		Company:		Variant:	Git Hash:
			NA		CHECKED	85c4a65
	Board Name:				Project Name:	
	Step-Up				Isolated step-up 12:450 V	
Sheet Title:	File Name:	Designer:	Date:	Revision:		
B.Mask (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-07	2.0.0+ (Unreleased)		
Sheet Path:		Reviewer:	Size:	Sheet:		
		FR	A4	8 of 10		



 ChurrasCod Electronics	Comments:		Company: NA		Variant: CHECKED	Git Hash: 85c4a65
	Sheet Title: F.Silkscreen (Scale 1:1)		Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
	Sheet Path:		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
			Reviewer: FR	Size: A4	Sheet: 9 of 10	



ChurrasCod
Electronics

Comments:

Company:

NA

Variant:

CHECKED

Git Hash:

85c4a65

Board Name:

Step-Up

Project Name:

Isolated step-up 12:450 V

Sheet Title:

B.Silkscreen (Scale 1:1)

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Date:

2026-02-07

Revision:

2.0.0+ (Unreleased)

Sheet Path:

Reviewer:

FR

Size:

A4

Sheet:

10 of 10